Generative AI: Video 1 Notes

Introduction: GenAI vs. ML vs. DL and why to learn it in 2026

Series: Complete GenAI series with Python and LangChain

Note: some words in the auto-generated transcript were garbled. Corrected from context: "Lunchan" → LangChain, "GK" → Groq, "VO" → Veo, "Z AI" → GenAI.

1. Introduction and Motivation

- Over the last two years, many GenAI applications have launched, such as **ChatGPT, Google Gemini, Claude, ElevenLabs, Groq**.
- Companies are adopting AI, and AI skills are now mentioned in almost every job discussion.
- Students are building projects with AI, and developers are integrating it to improve productivity.
- Common questions this video answers: (1) What is GenAI? (2) How do Machine Learning, Deep Learning and Generative AI differ? (3) What is the difference between Traditional AI and GenAI? (4) Why learn GenAI in 2026?
- **Key framing:** ML and DL are today's *traditional AI*, while GenAI is *modern AI*.

2. What is Generative AI?

Definition: A technology or system that, based on a user's query, **generates new content**.

- Answers are **not predefined**. Each time, you get freshly generated content.
- Content is **not limited to text**. It can be text, images, videos, audio and programming code.
- This ability to *generate* is why it is called Generative AI.

3. Traditional AI vs. Generative AI

3.1 Machine Learning (ML)

Example: Netflix recommendations

- Netflix tracks what you watch: title, description, release year, genre, and how long you watched.
- Based on your past behavior and choices, it **predicts** other movies you might like.
- **Summary:** ML learns from past experiences and behavior, then predicts.

3.2 Deep Learning (DL)

Example: Face unlock on smartphones

- During setup, the phone captures many images of your face from different angles.
- It learns **complex patterns**: eye shape, nose structure, facial geometry, and more.
- Later, it recognizes you even with a cap, glasses or different lighting, and unlocks.
- **Summary:** DL is when a machine learns complex, deep patterns (deep architectures) from data such as images.

3.3 Generative AI

Example: ChatGPT

- Ask "What is GenAI in short?" and it generates a new answer.
- The answer is not fixed. Clicking "try again" gives a *different* answer to the same question.
- Behind the scenes, ChatGPT, Gemini and Claude run on **LLMs**.

4. Types of Generative Models

Model	Full Form	What it does	Example
LLM	Large Language Model	Trained on huge amounts of text data and generates text	ChatGPT, Gemini, Claude
LIM	Large Image Model	Generates images from a user prompt	Google Nano Banana
LVM	Large Vision Model	Generates images or videos from a prompt	Google Veo
LAM	Large Audio Model	Generates audio (instructor asks viewers to guess and comment)	Text-to-speech and voice tools such as ElevenLabs

5. Why Learn GenAI in 2026?

- GenAI is the **most in-demand technology** right now.
- **Currently:** it is an *optional* skill. Knowing it gives you an edge, but not knowing it is not a big problem yet.
- **In the coming years:** it will become a **must-have skill**. The instructor's claim is that without GenAI skills, getting an IT job will be difficult.
- The series is presented as a great opportunity to upskill.

6. Series Promise

- At least **10 projects** will be built during the series.
- The course goes **from basic to advanced**.
- The next video begins the actual GenAI content, with Python and LangChain.

Quick Revision Summary

- **ML:** learns from past behavior and predicts (Netflix recommendations).
- **DL:** learns complex patterns, often from images (face unlock).
- **GenAI:** generates *new* content such as text, images, video, audio and code (ChatGPT).
- **LLM** = text, **LIM** = images, **LVM** = images and video, **LAM** = audio.
- ML and DL are **traditional AI**, while GenAI is **modern AI**.